

TED(10)-5053

Reg no.....

(Revision-2010)

Signature.....

Sixth Semester Diploma Examination in Engineering/Technology March 2013

Model Question Paper

Optical communication

(maximum marks:100)

PART-A

I answer the following questions in one or two sentences. Each question carries 2 marks.

- 1 What is mean by dispersion?
- 2 What is the difference between step index and graded index fibers?
- 3 Name any two optical switching networks?
- 4 Name the optical sources used in fiber optics communication?
- 5 What is optical isolator?

(5x2=10 Marks)

PART –B

II Answer any five of the following. Each question carries 6 marks.

- 1 Draw the block diagram of fiber optic communication system and explain?
- 2 Write short notes on meridional rays and skew rays?
- 3 Explain the principle of photo detection?
- 4 Explain the theory behind the laser action?
- 5 Draw the block diagram of optical transmitter?
- 6 List different types of optical couplers?
- 7 Explain the structure of packet switching network?

(5x6=30 Marks)

PART-C

(Answer any one full question from each unit)

UNIT-1

III a. Explain the attenuation in optical fibers? (15marks)

OR

IV a. Explain different types of fibers based on their refractive index and number of modes?
(15 marks)

UNIT -2

V a. Explain the laser diode structure? (8 marks)

b. Draw the structure of PIN Photo diode and explain (7 marks)

OR

VI a. Explain Erbium doped fiber amplifiers? (8marks)

b. Draw the structure of avalanche photo diode and explain its operations? (7 marks)

UNIT -3

VII a. Draw the block diagram of optical receiver and explain it? (8marks)

b. Write the working principle of optical circulators with suitable diagrams (7marks)

OR

VIII a. Explain wave length division multiplexing (8marks)

b. What is fiber splicing? Explain different fiber connectors? (7marks)

UNIT-4

IX a. Explain different network topologies used for fiber optic communication? (15marks)

OR

X a. Compare burst switching with circuit switching? (8marks)

B .Draw the block diagram of optical Ethernet and explain? (7marks)